
(a)
Where the VN+Huber gain lives: per-element MAE change vs ALIGNN, MP labels
n < 30 test structures (not scored)
element absent from the test split
H
He
Li
Be
B
C
N
O
F
Ne
Na
Mg
Al
Si
P
S
Cl
Ar
K
Ca
Sc
Ti
V
Cr
Mn
Fe
Co
Ni
Cu
Zn
Ga
Ge
As
Se
Br
Kr
Rb
Sr
Y
Zr
Nb
Mo
Tc
Ru
Rh
Pd
Ag
Cd
In
Sn
Sb
Te
I
Xe
Cs
Ba
Hf
Ta
W
Re
Os
Ir
Pt
Au
Hg
Tl
Pb
Bi
Po
At
Rn
Fr
Ra
La
Ce
Pr
Nd
Pm
Sm
Eu
Gd
Tb
Dy
Ho
Er
Tm
Yb
Lu
Ac
Th
Pa
U
Np
Pu
Am
Cm
Bk
Cf
Es
Fm
Md
No
Lr
−0.04044
−0.02022
0.00000
0.02022
0.04044
ΔMAE = (VN+Huber) − ALIGNN (eV); blue = improvement, white = no change
(b)
MAE by crystal system (symmetry ascending), MP labels
ALIGNN
ALIGNN+VN
ALIGNN+VN+Huber
Test MAE (eV)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
triclinic n=1,167
monoclinic n=2,218
orthorhombic n=1,791
tetragonal n=903
trigonal n=721
hexagonal n=552
cubic n=1,097
(c)
MAE by true band-gap bucket
Test MAE (eV)
0.10
0.15
0.20
0.25
0.30
0.35
0.40
ALIGNN
ALIGNN+VN
ALIGNN+VN+Huber
MP labels (filled)
HZW labels (open)
0 (metal)
(0,0.5]
(0.5,1]
(1,2]
(2,3]
(3,4]
>4
True band gap (eV)